Deependra Singh

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PhD Research Scholar in Physics
Harish-Chandra Research Institute (HRI), Prayagraj
LinkedIn Profile   GitHub

Education

2025 – Present — **PhD in Physics, Harish-Chandra Research Institute (HRI), Prayagraj**.
Research Area: *Quantum Information Theory and Quantum Foundations*.

Aug 2020 – Jun 2025 — **Integrated M.Sc. in Physics, NISER Bhubaneswar (HBNI)**.
Cumulative GPA: *8.58 / 10.00*.
Physics Core GPA: *8.55 / 10.00*.
Minor in Computer Science.

2020 — **Higher Secondary (ISC)**, St. John's School, Marhauli, Varanasi.
Percentage: *96.4%*.

2018 — **Secondary School Certificate (ICSE)**, St. John's School, Marhauli, Varanasi.
Percentage: *94.0%*.

Research Experience

May 2024 – May 2025 — **Master's Thesis Project, NISER Bhubaneswar**.
The Bosonic String in Flat Spacetime: Quantization and Conformal Field Structure.
Studied the quantization framework for bosonic strings in flat space. Involved detailed reading and analysis of conformal field theory, representation theory, non-Abelian gauge theories, and renormalization group flows.

May 2023 – July 2023 — **Summer Research Fellow, IISER Mohali**.
Evaluation of 1-Loop Feynman Integrals in Asymptotic Limits.
Derived asymptotic expansions in high- and low-mass limits for loop integrals (bubble, triangle, square) relevant to QED scattering processes. Performed analytical and numerical analysis using the `FeynCalc` package in Mathematica.

May 2022 – July 2022 — **Summer Research Fellow, IIT Bhilai**.
Physical Interpretation of Pressure inside Proton.
Developed models using classical and quantum statistical mechanics to explain experimental data on proton pressure. Acknowledged for the paper *"Proton Pressure in Terms of Degenerate Quark Distribution,"* presented at the *DAE Symposium on Nuclear Physics, Vol. 67 (2023)*.

2023–2025 — **Publications and Submissions**.
- *"Retinal Fundus Multi-Disease Image Classification using Hybrid CNN, Transformer, Ensemble Architecture," Lecture Notes in Networks and Systems, Springer (2023)*. DOI: 10.1007/978-981-97-7190-5_9.
- *"Interpretable AI Models for Detecting and Classifying Multiple Retinal Conditions," ACM Transactions on Computing for Healthcare* (under review, 2024).
- *"HetGSMOTE: Addressing Class Imbalance in Heterogeneous Graph Neural Networks,"* submitted to *NLDL 2026*.

2024 — **Teaching Assistant, Advanced Machine Learning Course (NISER)**.
Delivered lectures on advanced topics in deep learning and mentored two student project groups.

2023 — **Editorial Position, Inventa Science Magazine**.
Collaborative science magazine led by UG and PG students of IISERs, IISc, NISER, and CEBS, promoting interdisciplinary science communication.

Research Interests

- **Quantum Foundations and Quantum Information Theory:** Entanglement Theory, Measurement Problem, Collapse Models, Quantum Reference Frames, Open Systems, Quantum Thermodynamics, Quantum Causality, Quantum Gravity, Mathematical Quantum Foundations.
- **Interdisciplinary:** Machine Learning (Deep Learning, GNNs, PINNs, Topological Neural Networks, Explainability, Quantum AI), Computational Physics (Monte Carlo Methods, Simulations, Quantum Computing), Applications in Medical Science.

Selected Coursework Projects

- **Quantum Information and Computing:** Implemented BB84 and E91 Quantum Key Distribution protocols using Qiskit and SSH. `GitHub: QKD_Protocols`
- **Many-Body Physics:** Performed a Monte Carlo study of the Double Exchange Model under large Hund's coupling. `GitHub: Double-Exchange-Model`
- **Computational Physics:** Developed a Variational Monte Carlo (VMC) importance sampling framework for quantum ground-state estimation. `GitHub: VMC Importance Sampling`
- **Machine Learning:** Conducted a solar capacity impact correlation study using regression and feature importance analysis.

Skills

Languages	English (Full Professional Proficiency), Hindi (Native), and Elementary German.
Programming & Tools	Python (NumPy, SciPy, Matplotlib, SymPy, Pandas), C, Mathematica, MATLAB, and LaTeX.
Quantum & Physics Software	Qiskit, QuTiP, and familiarity with ROOT for particle physics analysis.
Machine Learning	PyTorch, PyTorch Geometric, scikit-learn; experience with deep learning architectures (CNNs, GNNs, Transformers), quantum machine learning, and explainable AI.

Achievements

Awards and Distinctions

2020	**JEE Advanced:** All India Rank 4026 (Top 0.4%).
	NEST: All India Rank 150 (Top 1%).
2022	**National Graduate Physics Examination:** National Topper (Top 1% all India).
2023	**GATE Physics:** All India Rank 617 (Top 5%).
2024	**CSIR-NET Physics:** Qualified for PhD (Top 5% all India). **TOEFL iBT:** Scored 116/120 (C2 Level Proficiency).
2025	**GATE Physics:** All India Rank 143 (Top 0.5%).
	CSIR-NET Physics: Qualified for JRF (Rank 126, Top 0.4% all India).
	JEST Physics: All India Rank 165 (Top 5%).
	PhD Selection Interviews: Cleared for BARC, SINP, and HRI, India.

Scholarships and Fellowships

2020–2025	**DISHA Scholar:** Awarded by the Department of Atomic Energy, India.
2025–Present	**Junior Research Fellowship (JRF):** Awarded for pursuing PhD in India.